INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

COMBINATION OVEN, ELECTRIC

The General Services Administration has authorized the use of this commercial item description (CID), for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers single-compartment and two-compartment combination ovens for use in the forced convection, steam, combined steam and convection modes.

1.2 Intended use. The combination oven is intended for use in Naval surface shipboard galleys.

2. CLASSIFICATION.

2.1 Classification. The combi-oven shall conform to the following types:

Type 1: Single compartment
Type 2: Two compartment(stacked)
Type 3: Single compartment(with door window)
Type 4: Two compartment(with door window)

Beneficial comments (recommendations, additions, deletions) and any data which may be of use in improving this document should be addressed to: U. S. Army Soldier Systems Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal DD Form (DD 1426) appearing at the end of this document or by letter.

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3. SALIENT CHARACTERISTICS

3.1 Design and construction. The combination ovens shall consist of a water tight insulated oven cavity, sealing center split doors, blower, air heater(s), steam generator, single exit discharge drain connection, hose washable cooking cavity with center drain; over pressure and under pressure venting, as required. Each oven compartment, when used in the steaming or combination steaming convection mode, shall use five gallons of water per hour or less, and be capable of operating independently with separate controls. All four oven types shall be fabricated from stainless steel with corrosion-resistant hardware and fittings in accordance with the manufacturer's standard practice. All threaded parts shall conform to ANSI B1.1.

The type 1 oven shall provide the following features:

- Single oven cavity
- Adjustable solid center split doors with vertical handles and interlock switch to shut off the fan, and heating elements when door is opened, except in cool down mode
- Five equally spaced removable stainless steel wire racks, capable of supporting a load of 20 pounds per square foot each
- Rail guides capable of being removed without the use of tools.
- Two-speed fan to provide circulation of forced heated air uniformly throughout the oven compartment
- Fan rotation clearly marked on the back oven baffle
- Fan selector switch on the control panel with a minimum speed of 1100 rpm when set at low speed and 1750 rpm when set at high speed
- Timer with 120-minute range, stay on mode and audible buzzer when time expires and stays on until manually shut off

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- Internal mounted, removable meat probe with stainless steel protection cable, controls that will turn off the oven when product temperature reaches the set point

- Internal oven bracket to stow the meat probe when not in use

- Wash down hose for interior cleaning, with a storage bracket, one per oven type

- Capability to operate on shore or distilled ship water

- Vertical slide-out electronic control panel with serviceability from the front

- A deliming kit with all required accessories supplied, one per oven type

The type 2 oven (includes the same features as type 1) shall consist of two identical ovens stacked and secured together.

The Type 3 oven (includes the same features as Type 1) shall be equipped with a window in one door.

The Type 4 oven (includes the same features as Type 1) shall consist of two identical ovens stacked and secured together and shall be equipped with a window in one door of each oven.

3.2 Dimensions. The ovens shall have the following maximum outside dimensions:

Type	Width (inches)	Height (inches)	Depth * (inches)
1	38 1/8	56	47
2	38 1/8	65	47
3	38 1/8	56	47
4	38 1/8	65	47

* Includes handles, drains and vents

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3.3 THERMAL PERFORMANCE

3.3.1 Convection mode performance. The oven temperature, starting from ambient air temperature of 70^{0}F, shall heat to 300^{0}F, 350^{0}F and 400^{0}F for hot air mode, when tested as specified in 5.2.1.

3.3.2 Steam mode performance. Each oven compartment shall be capable of performing in the steam mode, when tested as specified in 5.2.2 and 5.2.3.

3.4 Oven capacity. Each oven section shall have the capacity to accommodate the following pan sizes and quantities: ten steam table pans(12 inches by 20 inches by 2-1/2 inches high), or five baking pans(18 inches by 26 inches by 1 inch high), or two roasting pans (24 inches by 18 inches by 4-19/32 inches high). Pans shall not contact the interior oven wall surfaces.

3.5 Doors. Oven doors shall be stainless steel. The oven door(s) shall be insulated, with replaceable gaskets, and provision made to prevent condensate from dripping on the deck. The doors shall not require more than 16 lbs of force to unlatch during operation, when tested as specified in 5.2.4.

3.6 Types 3 and 4 doors. The doors for Types 3 and 4 ovens shall contain a clear tempered glass window that shall conform to ANSI Z97.1. The window shall be centered in the door a maximum size of 6 inches by 12 inches. The gasket shall meet NFS International Standard 2 for materials used in the food contact area. A shield shall be provided on the outside of the glass to prevent hand contact.

3.7 Oven door latch. Oven doors shall be equipped with secondary exterior mechanical latches (in addition to the main door latches). Each side-opening-type door shall be provided with a hold-open latch which holds the doors in a wide open position flush with the oven interior side walls.

3.8 Mounting. The type 1 oven shall be provided with four stainless steel legs not less than 18 inches high suitable for attachment to the deck. The type 2 oven shall be provided with either a separate oven base or four legs 6 ±1/2 inch long for bolting or welding to the deck. Bases and legs shall be fabricated from series 300 stainless steel, adequately reinforced.

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3.9 Inclined operation and structural integrity (shipboard requirement, see 7.2). The oven shall operate satisfactorily, in accordance with the requirements of this CID (such as the door remaining closed to prevent the loss of contents), when test-operated for 30 seconds inclined at an angle of 15 degrees each side of the vertical, in each of two vertical planes at right angles to each other. The oven shall not deviate or rack from perpendicularity to the deck by more than 1/8 inch when the deck is inclined at an angle of 15 ^{0} degrees.

3.10 Environmental suitability . Combination ovens shall be capable of withstanding ship's vibration in accordance with MIL-STD-167/1, type 1 environmental vibration without malfunction.

3.11 Temperature control. Each oven shall be equipped with an automatic adjustable thermostat to provide automatic heat control of the oven compartment and permit cooking operations to be accomplished by maintained temperature. The thermostat shall be capable of maintaining any oven temperature within the range of 150^{0}F to at least 450^{0}F, plus or minus 10^{0}F of the temperature setting.

3.12 Heating units. The oven shall be provided with heating elements located to provide even heat distribution throughout the oven compartment on all operating modes. The heating elements shall be fully protected from the pans and fan blade, and removable from the oven front.

3.13 Heating elements. The ends of the heating elements shall be sealed and protected to preclude the entrance of moisture. Heating elements shall be removable from within the oven cavity. Heating element leads requiring disconnection for removal of the elements shall be accessible from inside the cavity.

3.14 Surface temperature. Each oven unit shall be capable of operating up to 450^{0}F with no external contact surface exceeding a temperature of 140^{0}F, unless guarded or clearly marked as specified in 3.25.

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3.15 Electrical requirements. The units shall operate on 440 VAC, 60 Hz, three-phase power, three wire with 120 volt controls as defined in MIL-STD-1399, section 300, and shall have provisions for making direct (hardwired) connections for electrical power (2 conductors) and for equipment grounding (1 conductor). All outermost metallic surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be not greater than 0.1 ohm. Electrical components, other than hermetically sealed motor, shall be provided in accordance with NEMA 250, type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45.

3.16 Electromagnetic compatibility. When required, each unit type shall be designed for electromagnetic compatibility in accordance with the requirements of MIL-STD-461, class A4 for surface ships.

3.17 Electrical wiring. Electrical wiring shall be arranged, supported and marked to allow identification with provided wiring diagram for each unit.

3.18 Controls. All controls, switches, and indicator lights shall be readily accessible and easily operable for steam, hot air and combination modes; and shall be located in a waterproof control panel at the front right side of the door.

3.19 Accessibility. The unit shall be physically and visually accessible (including motor assembly, fans, heating elements, steam generator, and controls) from the front of the oven for operation and for normal maintenance with tools, test equipment, and replacement parts. No special tools shall be required for access or parts replacement.

3.20 Water and electrical connections. Each unit shall be equipped with a water and electrical hook-up connection located at the front of the unit. The water inlet shall be clearly marked with water pressure requirements. The water and electrical connections shall not interfere with the operation of the oven.

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3.21 Physical size limitations. The units shall pass or shall disassemble without any need of cutting and/or modification, so that it can pass through a shipboard access measuring 26 inches wide by 66 inches high with 8 inch radius corners.

3.22 Label plates. The unit shall be provided with a data nameplate and an instruction plate, both attached to the front doors of the unit. They shall be readily visible during normal operating use and shall not adversely affect the life and utility of the unit.

3.23 Data nameplate. The data nameplate shall contain the manufacturer's name, model, serial number, date manufactured, and other information needed to uniquely identify the unit.

3.24 Instruction plate, and operation manual. The unit shall be provided with an instruction plate affixed to the right hand door, and an operations manual for start-up, operation, shut down, and cleaning (procedures for deliming).

3.25 Hot spot labels. Each unit shall be clearly marked with labels indicating hot external spots(see 3.14). Labels shall be capable of withstanding surface temperatures, cleaning products, and the cleaning process without degradation.

3.26 High voltage labels. A label reading "Danger-High Voltage" shall be affixed to the outer case assembly, on or adjacent to each service access cover near one of the fasteners securing the cover. In addition, a warning label in accordance with UL 969 shall be placed near the high voltage components inside the equipment. This label shall include, but not be limited to the following texts:

- Danger-High Voltage
- Power supply must be disconnected before servicing
- Access covers must be in place before use
- Service should be performed by authorized personnel only

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3.27 Standard compliance. The unit shall be in accordance with and certified to meet the applicable requirements of NSF 4 and UL 197.

3.28 Workmanship. The finished units shall be free of cracks, dents, or deep scratches. All burrs, slivers, splinters, sharp edges, dirt, grease, and foreign matter shall be removed in both easy access areas and where access panels are removed.

4. REGULATORY REQUIREMENT

4.1 Contractor recovered material. The offeror/contractor is encouraged to use recovered materials to maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and/or be the same product that has successfully been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such compliance to all salient characteristics.

5.2 TESTING

5.2.1 Convection mode performance test. The oven cavity shall heat to 300^{O}F in 10 minutes or less at an ambient temperature of 70^{O}F plus or minus 5^{O}F. The oven shall be set at successively higher temperatures of 350^{O}F and 400^{O}F, the oven compartment temperature shall be measured. The temperature shall not vary more than plus or minus 10 ^{O}F from the oven set temperature. The oven temperatures shall be recorded to determine compliance with 3.3.1.

5.2.2 Steam mode performance test. With the oven in the steam mode four-dozen large eggs shall be loaded in the oven cavity and held for 13 minutes. Three eggs selected at random shall be opened with in one minute to determine compliance with 3.3.2. All sample eggs shall exhibit a solid yoke. When a double-stacked oven is tested, both cavities shall be tested for steam performance at the same time.

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5.2.3 Steam mode capacity test. Fifty pounds of water shall be equally divided into six, four inch deep steam table pans at a temperature of 70^{0}F plus or minus 5^{0}F. The pans shall be placed in the oven, the oven turned on to the steam mode. The water shall reach 200^{0}F in one hour or less to determine compliance with 3.3.2.

5.2.4 Door pull test. The oven door shall not require more than 16 pounds of force to open when operating at 450^{0}F to determine compliance with 3.5.

6. PACKAGING

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Operation and repair manual, inspection and testing reports. Each unit shall be supplied with an inspection and testing report; and an operation and repair manual meeting the requirements of ASTM F760.

7.2 Ordering data. Purchasers should include the following information in procurement documents.

a. Title, number and date of this document.
b. Type required (see 2.1 and 3.16).
c. Packaging requirements(see 6.1).

7.3 Sources of document.

7.3.1 Sources of government documents.

MILITARY STANDARDS:

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally excited)

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements For The Control of Electromagnetic Interference

MIL-STD-1399 - Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating Current

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(Unless otherwise indicated, copies of the above standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

7.3.2 Sources for non-government association documents. Copies of referenced non-government documents may be obtained from the following organizations:

AMERICAN NATIONAL STANDARDS INSTITUTE:

ANSI B1.1 – Unified Inch Screw Threads(UN & UNR Thread Form)

ANSI Z97.1 - Glazing Materials used in Buildings - Safety Performance Specifications and Methods of Test

(Unless otherwise indicated, copies of the above standard is available from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036).

AMERICAN SOCIETY FOR TESTING AND MATERIALS:

F760 – Standard Specification for Food Service Equipment Manuals

(Unless otherwise indicated, copies of the above standard is available from the ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC.:

IEEE 45 – IEEE Recommended Practice for Electric Installation on Shipboard

(Unless otherwise indicated, copies of the above standard is available from the Institute of Electrical and Electronics Engineer, Inc., 445 Hoes Lane, Piscataway, NJ 08855-1331).

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION:

NEMA 250 – Enclosures for Electrical Equipment
(1000 Volts Maximum)

(Unless otherwise indicated, copies of the above standard is available from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosalyn, VA 22209).

NSF INTERNATIONAL:

NSF 2 – Food Service Equipment

NSF 4 – Commercial Cooking, Rethermalization and Powered Hot Food Holding and Transport Equipment

UNDERWRITERS LABORATORIES INC.:

UL 197 – Commercial Electric Cooking Appliances
UL 969 – Marking and Labeling Systems

(Unless otherwise indicated, copies of the above standards are available from the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062).

7.4 Sources of supply. A supplier who is known to manufacture a combi-oven is listed below; however, competition is not limited to these companies.

G.S. Blodgett Company
55 Boyer Circle
P.O. Box 1440
Williston, VT 05495-1440

Market Forge Industries Inc.
35 Garvey Street
Everett, MA 02149-4403

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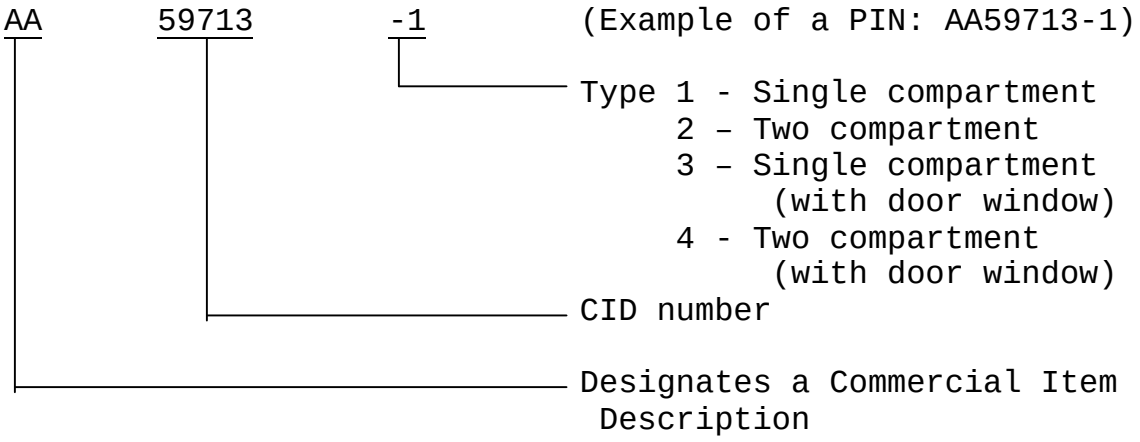
7.5 Part identification number(PIN). The PINs to be used for items required by this CID are created as follows:



MILITARY INTERESTS:

Custodians
Army – GL
Navy – SA

Review Activities
Army –MD1
Navy – SH
DLA – SS

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA-FSS/7FXE

Preparing Activity:

Army - GL

DoD Project (7310-0946)

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